

Burns and scalds: first aid

A burn is damage to the skin from heat, hot liquids (a "scald"), electricity, chemicals, or radiation such as the sun. What you do in the first minutes matters. One of the most useful first actions for most burns is simple: cool it with running water, and do not put creams, oils, or home remedies on the fresh burn as first aid unless a clinician or local guidance tells you to.

CALL EMERGENCY SERVICES NOW

Get emergency help for a serious burn

Call your local emergency number (911 in the US/Canada, 112 in much of Europe, 999 in the UK) for a burn that is large (bigger than the person's hand), deep, or on the face, hands, feet, genitals, or over a joint; any burn that goes all the way around a limb; any electrical or chemical burn; any burn with trouble breathing, coughing, a hoarse voice, or soot around the nose or mouth (the airway may be burned); and any serious burn in a baby, young child, pregnant person, or older or frail person. **While you wait for help, start cooling the burn.**

What to do, step by step

- 1 Stop the burning and stay safe.** Move the person away from the heat source. If clothing is on fire, **stop, drop, and roll**, or smother the flames with a blanket. Switch off the power before touching someone with an electrical burn. For a chemical burn, brush off any dry chemical and remove contaminated clothing, protecting your own skin.
 - 2 Cool the burn with running water for 20 minutes.** Hold the burned area under cool or lukewarm *running* water as soon as possible. Keep cooling for a full 20 minutes — it still helps even up to about 3 hours after the burn. Do **not** use ice, iced water, or very cold water, which can deepen the injury. If there is no running water, use any clean, cool water you can pour over it.
 - 3 Remove jewellery and tight or soaked clothing** from near the burn before it starts to swell — but do not pull off anything that is stuck to the skin.
 - 4 Keep the rest of the person warm.** Cooling a large burn can quickly chill the body, especially in children and older people. Cover the parts you are not cooling with a blanket or coat, and avoid cooling so much that they start to shiver.
 - 5 Cover the burn loosely** once it is cooled. A layer of clean kitchen cling film (plastic wrap) laid over the burn is ideal — lay it on lengthwise, do not wrap it tightly around a limb. A clean, non-fluffy cloth or a clean plastic bag (good for a hand or foot) also works. This eases pain and keeps the wound clean.
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DO NOT USE HOME REMEDIES ON A FRESH BURN

Do **not** apply butter, oil, lard, toothpaste, egg, soy sauce, ash, soil, cow dung, herbs, ice, or "burn creams" and ointments to a fresh burn as first aid unless a clinician or local guidance specifically tells you to. These can trap heat, raise the risk of infection, and make it harder for medical staff to see and treat the burn. **Do not burst blisters** — they protect the skin underneath. Cool water and a clean loose cover are usually the safest first-aid choice.

Chemical and electrical burns

Chemical burns

- Protect yourself first — wear gloves or use a barrier if you can.
- Brush off any **dry** powder, then rinse the area with lots of cool running water for at least 20 minutes, unless the product label, poison centre, or emergency guidance tells you not to use water.
- Remove clothing and jewellery the chemical has touched while rinsing.
- If a chemical is in the eye, rinse the eye gently with clean water for at least 20 minutes and get medical help.
- If you know the chemical, tell the medical team or poison centre.

Electrical burns

- **Do not touch the person** until you are sure the power is switched off — you can be electrocuted too.
- Electrical burns can be far deeper than they look on the surface and can affect the heart.
- Always seek medical care, even if the skin damage seems small.
- If the person is unresponsive and not breathing normally, call for help and start CPR.

Smaller burns you can treat at home

A small, shallow burn — red and sore, perhaps with small blisters, no bigger than the person's hand and not on the face, hands, feet, genitals, or a joint — can usually be cared for at home: cool it, cover it loosely, and keep it clean. Simple pain relief such as paracetamol/acetaminophen or ibuprofen can help if safe for that person (follow the label and any local guidance). **Get medical advice if** the burn becomes more painful, swollen, smelly, or oozing, or the person develops a fever — these can be signs of infection. Make sure tetanus protection is up to date for any burn that breaks the skin.

Babies, children, and others at higher risk

LOWER THE THRESHOLD TO GET HELP

Babies and young children, older or frail people, and pregnant women are more easily harmed by burns. Get medical advice for any burn that blisters in a baby or child, and don't wait if you are unsure. Children also chill quickly when a burn is cooled, so keep the rest of the body warm.

Preventing burns and scalds

- Keep hot drinks, pots, and kettles out of children's reach; turn pot handles inward; use the back rings of the stove.
- If you control the water heater, set it to a safer temperature (about 50 °C / 120 °F) and test bath water before a child gets in.
- Keep cooking fires and stoves stable and away from children; keep flammable clothing away from open flames.
- Fit smoke alarms and check them; keep matches, lighters, and chemicals locked away.
- Protect skin from the sun and avoid sunburn.

Sources

- World Health Organization — Burns fact sheet (global burden, prevention, and first aid: cool with running water, do not apply harmful substances).
- American Red Cross — first-aid training and resources (burn first aid and when to seek care).
- American Heart Association — first-aid guidelines (cooling burns; thermal, chemical, and electrical injuries).
- NHS — Burns and scalds (cool 20 minutes, cover with cling film, do not burst blisters, when to get help).

Guidance reflects widely taught burn first aid for lay responders. Details vary between training bodies and regions — follow your local first-aid training and emergency services.

This guide is general information, not medical advice. In an emergency, always call your local emergency number. Reuse and reprint freely.